

The ± 1 Invariant: Parity as a Multiplicative Tool

PRELIMINARIES

An invariant is a quantity preserved under the operations of a system. Among the simplest and most useful invariants is parity, which works very efficiently when a problem requires translation from an additive to a multiplicative setting.

The map $x \mapsto (-1)^x$ sends even integers to 1 and odd integers to -1 , converting statements about parity into statements about products in $\{+1, -1\}$. This translation is valuable because multiplication interacts cleanly with structural features of a problem: squares vanish, permutations do not affect products, and repeated factors collapse. The recurring strategy in what follows is a *double count* of a single quantity P . One evaluation uses the local definition of P (what each factor is, combinatorially); the other uses a global property (a count of negative entries/ a symmetry/ a cancellation). Equating the two, we can arrive at a divisibility or parity conclusion that was not visible from either side alone.

A useful heuristic: whenever a problem gives a condition of the form “a certain sum of ± 1 ’s equals zero,” that condition is really telling you *how many negatives there are*. Whenever the conclusion involves divisibility by 2 or 4, parity in the multiplicative operations is a natural first tool to try.

PROBLEM 1. SIGNS ON A POLYGON

Statement. Assign to each vertex of a regular n -gon ($n \geq 3$) a value in $\{1, -1\}$. On each edge write the product of the values at its two endpoints. If the sum of the edge values is 0, prove that $4 \mid n$.

Discussion. The condition $\sum e_i = 0$ does not provide much information: it seems to say only that the edges are balanced. But since each $e_i \in \{1, -1\}$, a zero sum is extremely rigid — it forces exactly $n/2$ edges to be -1 and $n/2$ to be $+1$. This immediately gives $2 \mid n$, which is halfway to the goal. To reach $4 \mid n$, we need to extract one more bit of parity information, and the hypothesis has already been fully used. Where does the extra structure come from?

The answer is that we have not yet used the fact that the e_i are not arbitrary ± 1 ’s: they are defined as products $v_i v_{i+1}$ of vertex labels. This definition introduces a hidden constraint, namely that each vertex appears in exactly two edges. The technique for exposing such hidden constraints is to form the product $P = \prod e_i$ and evaluate it in two ways: once using the edge *values* (which gives $(-1)^k$ where k is the number of negative edges), and once using the edge *definitions* (which collapses because each v_i appears twice, and $v_i^2 = 1$). The two evaluations must agree, and this forces the parity of k .

Proof. Label the vertices $v_1, v_2, \dots, v_n \in \{1, -1\}$ cyclically, with the convention $v_{n+1} = v_1$. For $1 \leq i \leq n$, let $e_i = v_i v_{i+1}$ denote the edge value.

Since each $e_i \in \{1, -1\}$, write k for the number of indices with $e_i = -1$. The condition $\sum_{i=1}^n e_i = 0$ gives

$$(n - k)(1) + k(-1) = 0, \quad \text{so } n = 2k.$$

Consider the product $P = \prod_{i=1}^n e_i$. We evaluate P in two ways.

First evaluation. Since P is a product of $n - k$ factors equal to 1 and k factors equal to -1 ,

$$P = (-1)^k.$$

Second evaluation. Substituting $e_i = v_i v_{i+1}$,

$$P = \prod_{i=1}^n v_i v_{i+1} = (v_1 v_2)(v_2 v_3) \cdots (v_n v_1).$$

Each vertex v_j occurs exactly twice in this product (once as the right endpoint of e_{j-1} and once as the left endpoint of e_j). Hence

$$P = \prod_{j=1}^n v_j^2 = 1,$$

since $v_j^2 = 1$ for every j .

Comparing the two evaluations, $(-1)^k = 1$, so k is even. Writing $k = 2m$, we get $n = 2k = 4m$. Therefore $4 \mid n$. □

PROBLEM 2. A SLIDING WINDOW OF LENGTH FOUR

Statement. Let $a_1, \dots, a_n \in \{1, -1\}$ be arranged on a circle. For each i , let $S_i = a_i a_{i+1} a_{i+2} a_{i+3}$ (indices modulo n). If $\sum_{i=1}^n S_i = 0$, prove that $4 \mid n$.

Discussion. This problem is the natural generalization of Problem 1: edges are replaced by length-four windows, and the underlying machinery is unchanged. The hypothesis $\sum S_i = 0$ again forces exactly half of the S_i to equal -1 , giving $n = 2k$. As before, we need an independent evaluation of the global product $P = \prod S_i$ to pin down the parity of k .

The key observation is combinatorial: when we expand P , every a_j appears with some multiplicity, and we need that multiplicity to be even (so the factors collapse to 1). Here the relevant count is how many windows contain a given index j . A window $S_i = a_i a_{i+1} a_{i+2} a_{i+3}$ contains a_j precisely when $i \in \{j - 3, j - 2, j - 1, j\}$, so each a_j appears in exactly four windows. Four is even, $a_j^4 = 1$, and the product collapses completely. Note that the same argument works for windows of *any* fixed even length ℓ , because each a_j then appears in exactly ℓ windows and $a_j^\ell = 1$. The structural content of the problem is that the window length is compatible with the parity we want.

Proof. All indices are taken modulo n . Each $S_i = a_i a_{i+1} a_{i+2} a_{i+3}$ lies in $\{1, -1\}$. Let k be the number of indices i with $S_i = -1$. From $\sum_{i=1}^n S_i = 0$,

$$(n - k)(1) + k(-1) = 0, \quad \text{so } n = 2k.$$

Define $P = \prod_{i=1}^n S_i$ and evaluate it in two ways.

First evaluation. Counting signs,

$$P = (-1)^k.$$

Second evaluation. Expanding by definition,

$$P = \prod_{i=1}^n a_i a_{i+1} a_{i+2} a_{i+3}.$$

We count the multiplicity of a fixed a_j in this product. The element a_j appears in S_i precisely when $j \in \{i, i+1, i+2, i+3\}$, i.e. when $i \in \{j-3, j-2, j-1, j\}$. Hence a_j appears in exactly four of the windows, so

$$P = \prod_{j=1}^n a_j^4 = 1,$$

since $a_j^4 = 1$ for $a_j \in \{1, -1\}$.

Equating the two evaluations, $(-1)^k = 1$, so k is even. Writing $k = 2m$ gives $n = 2k = 4m$, proving $4 \mid n$. $\square$

PROBLEM 3. ROW AND COLUMN SIGN-FLIPS

Statement. An 8×8 grid contains sixty-three entries equal to 1 and one entry equal to -1 . A move consists of multiplying every entry in a chosen row or column by -1 . Can the grid be reduced to all 1's?

Discussion. Reachability problems of this type are almost never solved by constructing an explicit sequence of moves; the move space is far too large. The standard approach is to look for an *invariant*: a function of the grid state that does not change under any permissible move. If we can find such a function that takes different values on the initial and target states, the target is unreachable.

Where should we look for an invariant? The operation acts on eight cells simultaneously by negating them. Any function built additively from individual cells will change in complicated ways, but the *product* of all entries behaves cleanly: negating eight entries multiplies their product by $(-1)^8 = 1$, and leaves the other 56 entries alone. So the product of all 64 entries is preserved exactly. Equivalently, the operation changes the number of -1 's by an even amount ($0, \pm 2, \pm 4, \pm 6$, or ± 8 , depending on how many of the eight flipped cells were already negative). The initial state has an odd number of -1 's; the target has zero. An odd count cannot become even under moves that preserve its parity.

Proof. We claim the required configuration is not reachable. To prove this, we establish an invariant that distinguishes the two states.

Let M denote the product of all 64 entries of the grid. In the initial configuration there are 63 entries equal to 1 and one entry equal to -1 , so

$$M_{\text{initial}} = 1^{63} \cdot (-1) = -1.$$

In the target configuration every entry equals 1, so $M_{\text{target}} = 1$.

We now show that M is unchanged by any permissible move. Suppose a move is applied to a row or column L , which contains exactly 8 entries. Let M_{old} and M_{new} be the values of M before and after the move. The entries outside L are unaffected, while each of the 8 entries of L is multiplied by -1 . Therefore

$$M_{\text{new}} = M_{\text{old}} \cdot (-1)^8 = M_{\text{old}}.$$

Hence M is invariant under every permissible move.

Since $M_{\text{initial}} = -1$ and $M_{\text{target}} = 1$, no sequence of moves can transform the initial grid into the target grid. $\square$

PROBLEM 4. AN ORTHOGONALITY CONDITION

Statement. Let $A = (a_{ij})$ be an $n \times n$ matrix ($n \geq 3$) with entries in $\{1, -1\}$ whose first column is all 1's. Suppose any two distinct columns are orthogonal. Prove that $4 \mid n$.

Discussion. This is the central lemma behind the existence theory of Hadamard matrices: ± 1 matrices with pairwise orthogonal columns. The conclusion $4 \mid n$ explains why such matrices exist only in sizes 1, 2, and multiples of 4. The hypothesis gives us a large amount of information — every pair of columns is orthogonal — but we will need only three of the $\binom{n}{2}$ relations, those among the first three columns.

A direct attack via case analysis of individual rows is hopeless. The trick is to find an expression whose *global* value equals n , but whose *local* values at each row are manifestly divisible by 4. Term-by-term divisibility then forces the conclusion. To construct such an expression, note that for $x \in \{1, -1\}$ the quantity $1 + x$ is either 0 or 2. A product of two such quantities therefore lies in $\{0, 4\}$ — and this $\{0, 4\}$ behaviour is exactly the local multiple-of-4 structure we want.

Which rows should we pick? The first column is all 1's, so $1 + a_{k1} = 2$ contributes nothing informative. We want expressions involving columns 2 and 3, chosen so that when we expand and sum, the linear and cross terms are killed by the orthogonality relations

$$\sum_k a_{k2} = 0, \quad \sum_k a_{k3} = 0, \quad \sum_k a_{k2}a_{k3} = 0,$$

leaving only the constant term, which sums to n . The expression $(1 + a_{k2})(1 + a_{k3})$ does exactly this.

Proof. Since the first column consists of 1's, the orthogonality of column 1 with column 2 states that $\sum_{k=1}^n a_{k1}a_{k2} = \sum_{k=1}^n a_{k2} = 0$. Similarly, orthogonality of column 1 with column 3 gives $\sum_{k=1}^n a_{k3} = 0$, and orthogonality of column 2 with column 3 gives $\sum_{k=1}^n a_{k2}a_{k3} = 0$. We record these three identities:

$$\sum_{k=1}^n a_{k2} = 0, \quad \sum_{k=1}^n a_{k3} = 0, \quad \sum_{k=1}^n a_{k2}a_{k3} = 0. \quad (*)$$

Define

$$S = \sum_{k=1}^n (1 + a_{k2})(1 + a_{k3})$$

and evaluate S in two ways.

First evaluation (global). Expanding the product inside the sum,

$$S = \sum_{k=1}^n (1 + a_{k2} + a_{k3} + a_{k2}a_{k3}).$$

By linearity and the identities $(*)$,

$$S = \sum_{k=1}^n 1 + \sum_{k=1}^n a_{k2} + \sum_{k=1}^n a_{k3} + \sum_{k=1}^n a_{k2}a_{k3} = n + 0 + 0 + 0 = n.$$

Second evaluation (local). Fix k . Since $a_{k2} \in \{1, -1\}$, the factor $1 + a_{k2}$ lies in $\{0, 2\}$; likewise $1 + a_{k3} \in \{0, 2\}$. Consequently,

$$(1 + a_{k2})(1 + a_{k3}) \in \{0, 4\}.$$

In particular, each summand is divisible by 4, hence so is S .

Combining the two evaluations, $n = S$ is divisible by 4. Therefore $4 \mid n$. □

PROBLEM 5. PERMUTATIONS AND ABSOLUTE DIFFERENCES

Statement. Let $a_1, \dots, a_n$ be integers and let $b_1, \dots, b_n$ be a permutation of them. Prove that

$$S = \sum_{i=1}^n |a_i - b_i|$$

is even.

Discussion. This is a parity statement about a sum, and the most efficient way to prove an integer S is even is to show $(-1)^S = 1$. Translating to the multiplicative domain has two advantages here. First, a sum inside an exponent becomes a product, which converts the problem from one about sums into one about products. Second, products are insensitive to the order of their factors, and the hypothesis we have — that (b_i) is a permutation of (a_i) — is precisely a statement about reordering.

There is one technical point. We want to manipulate $(-1)^{|a_i - b_i|}$, but the absolute value is awkward. The saving observation is that $|x|$ and x always have the same parity (since $|x| = \pm x$), so $(-1)^{|x|} = (-1)^x$. Once the absolute value is discarded, the exponent splits as $a_i - b_i$, the product factors into numerator and denominator, and the permutation hypothesis forces these two factors to be equal.

Proof. To show that S is even, it suffices to prove that $(-1)^S = 1$.

For any integer x , since $|x| = \pm x$, we have $|x| \equiv x \pmod{2}$, and therefore

$$(-1)^{|x|} = (-1)^x. \quad (*)$$

Now,

$$(-1)^S = (-1)^{\sum_{i=1}^n |a_i - b_i|} = \prod_{i=1}^n (-1)^{|a_i - b_i|}.$$

Applying $(*)$ to each factor,

$$(-1)^S = \prod_{i=1}^n (-1)^{a_i - b_i} = \frac{\prod_{i=1}^n (-1)^{a_i}}{\prod_{i=1}^n (-1)^{b_i}}.$$

Since $(b_1, \dots, b_n)$ is a permutation of $(a_1, \dots, a_n)$, the two multisets $\{a_1, \dots, a_n\}$ and $\{b_1, \dots, b_n\}$ are equal, and therefore

$$\prod_{i=1}^n (-1)^{b_i} = \prod_{i=1}^n (-1)^{a_i}.$$

The numerator and denominator are equal, so $(-1)^S = 1$. Hence S is even. $\square$